Zhengyang Yu

Software Engineering Student

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA
Expected Graduation: June 2028; GPA: 4.0/4.0
Relevant Coursework: Systems Programming, Computer Organization, Advanced Data Structures, Design and Analysis of Algorithms

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*
- Designed and built a C++/FreeRTOS application that authenticates multiple accounts, retrieves structured data over HTTPS, and presents live state through an interactive physical display.
- Implemented concurrent network workers, adaptive polling, configuration persistence, structured diagnostics, and bounded logs; debugged network, state-management, and rendering failures on hardware.
- Organized the firmware into maintainable service and UI modules, documented its custom server API, and used repeatable PlatformIO build and deployment workflows.

2025 **HTML to ePub Converter**, *Personal Project*
- Engineered a Python data-processing pipeline that parses HTML and XML, validates document structure, embeds metadata, and produces standards-compliant ePub files.
- Improved maintainability through modular transformations, structured error handling, and repeatable validation of generated output.

2024 **Home Assistant Presence Plugin**, *Personal Project*
- Built a Python integration that models room occupancy from entry and exit sensor events and exposes real-time state for home automation.
- Designed an event-driven REST API workflow with explicit state transitions to support testing, troubleshooting, and reliable smart-lighting behavior.

Skills

Programming	C++, Python, Java, C
Backend & Data	HTTP/HTTPS, REST APIs, JSON, XML, data processing, concurrent and event-driven systems
Engineering	Object-oriented design, data structures, algorithms, software testing, debugging, technical documentation
DevOps & Tools	Git branches and commits, Docker, Linux, PlatformIO, VS Code
Systems	FreeRTOS, structured logging, runtime diagnostics, configuration management

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch
Supported community members with computer troubleshooting and taught basic software skills.